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PII: S0021-9290(17)30146-X
DOI: http://dx.doi.org/10.1016/j.jbiomech.2017.03.004
Reference: BM 8155

To appear in: *Journal of Biomechanics*

Received Date: 26 August 2016
Revised Date: 6 March 2017
Accepted Date: 6 March 2017

Please cite this article as: Y.P. Lim, Y-C. Lin, M.G. Pandy, Effects of step length and step frequency on lower-limb muscle function in human gait, *Journal of Biomechanics* (2017), doi: http://dx.doi.org/10.1016/j.jbiomech.2017.03.004

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EFFECTS OF STEP LENGTH AND STEP FREQUENCY ON LOWER-LIMB MUSCLE FUNCTION IN HUMAN GAIT

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REVISION 2

Submitted to the Journal of Biomechanics

5 March 2017

Word count: 3,571 (Introduction to Discussion)

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ABSTRACT

The aim of this study was to quantify the effects of step length and step frequency on lower-limb muscle function in walking. Three-dimensional gait data were used in conjunction with musculoskeletal modelling techniques to evaluate muscle function over a range of walking speeds using prescribed combinations of step length and step frequency. The body was modelled as a 10-segment, 21-degree-of-freedom skeleton actuated by 54 muscle-tendon units. Lower-limb muscle forces were calculated using inverse dynamics and static optimization. We found that five muscles – GMAX, GMED, VAS, GAS, and SOL – dominated vertical support and forward progression independent of changes made to either step length or step frequency, and that, overall, changes in step length had a greater influence on lower-limb joint motion, net joint moments and muscle function than step frequency. Peak forces developed by the uniarticular hip and knee extensors, as well as the normalized fiber lengths at which these muscles developed their peak forces, correlated more closely with changes in step length than step frequency. Increasing step length resulted in larger contributions from the hip and knee extensors and smaller contributions from gravitational forces (limb posture) to vertical support. These results provide insight into why older people with weak hip and knee extensors walk more slowly by reducing step length rather than step frequency and also help to identify the key muscle groups that ought to be targeted in exercise programs designed to improve gait biomechanics in older adults.

Keywords: walking speed, cadence, support, progression, musculoskeletal model

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INTRODUCTION

Humans walk at different speeds by varying their step length and step frequency independently. The product of these two variables determines walking speed. Previous studies have investigated the effects of step length and step frequency on walking biomechanics by comparing the gait patterns of older adults against those of healthy controls. Kerrigan et al. (1998), Judge et al. (1996), Elble et al. (1991), and Winter et al. (1990) found that older adults walk more slowly than healthy controls by reducing their step length rather than step frequency. Older adults also walk with reduced peak hip extension angles (Kerrigan et al., 1998), decreased range of motion at the knee (Judge et al., 1996), smaller peak ankle plantarflexion angles (Judge et al., 1996; Kerrigan et al., 1998), and reduced peak hip-extension, knee-extension and ankle-plantarflexion moments (Judge et al., 1996; Kerrigan et al., 1998). Whilst these differences in joint kinematics and kinetics between young and older adults suggest that step length may have a greater effect on muscle function than step frequency, the results are confounded by the effect of age. Investigating a single cohort of healthy young adults would address this concern as it would more clearly delineate the separate effects of step length and step frequency on gait biomechanics and lower-limb muscle function.

The influence of walking speed on muscle-fiber lengths and muscle contributions to the ground reaction force in healthy young adults has been investigated using musculoskeletal modeling techniques. Arnold et al. (2013) found that a change in walking speed affects the region of the force-length curve on which the ankle plantarflexors operate, while the functional performance of the other lower-limb muscles appear to be less affected by walking speed. Previous studies also have shown that five muscles - gluteus maximus (GMAX), gluteus medius (GMED), vasti (VAS), gastrocnemius (GAS) and soleus (SOL) - contribute

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most substantially to the vertical and fore-aft components of the ground reaction force across a wide range of walking speeds (Jansen et al., 2013; John et al., 2012; Liu et al., 2008; Pandy and Andriacchi, 2010; Pandy et al., 2010). The results of these studies suggest that walking speed affects the magnitudes of the muscles' contributions to vertical support and forward progression more than their timing or coordination (Pandy and Andriacchi, 2010). No study to our knowledge has examined the separate effects of step length and step frequency on lower-limb muscle function during gait, specifically, the contributions of individual muscles to the vertical and fore-aft components of the ground reaction force and the region of the force-length relationship on which an individual muscle operates.

The aim of the present study was to quantify the separate effects of step length and step frequency on lower-limb muscle function during walking in healthy young adults. Three-dimensional gait data were used in conjunction with musculoskeletal computer modelling to evaluate muscle function as participants walked at a wide range of speeds using prescribed combinations of step length and step frequency. We evaluated the individual contributions of the major hip, knee and ankle extensors, specifically, GMAX, GMED, VAS, SOL and GAS, to vertical support and forward progression as step length and step frequency were changed independently. We hypothesized that muscle function would be more greatly affected by changes in step length than step frequency.

METHODS

Ten healthy young adults (age, 24.2 ± 2.7 years; height, 169.7 ± 9.3 cm; mass, 67.7 ± 10.5 kg; BMI, 23.5 ± 2.6 kg/m^2) were recruited to this study. Each participant provided written informed consent after approval was obtained from the University's Human Research

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Ethics Committee. All participants walked unassisted and were free from any pathology that could influence their ability to ambulate normally.

Gait experiments were conducted in the Biomotion Laboratory at the University of Melbourne. Each participant walked at six different speeds ranging from slow to fast using a prescribed combination of step length and step frequency at each speed. The range of walking speeds was achieved using three different step lengths – 80% (Short), 100% (Nominal) and 120% (Long) of a nominal value of step length – in combination with three different step frequencies – 80% (Slow), 100% (Nominal) and 120% (Fast) of a nominal value of cadence (Fig. 1). Note that only six different walking speeds can be derived from the nine prescribed combinations of step length and step frequency due to the commutative law of multiplication (e.g., 80% SL × 100% SF = 100% SL × 80% SF). The nominal values of step length and cadence were 0.73 m and 1.92 steps/sec, respectively, and were chosen based on data reported for healthy young adults (Cofre et al., 2011; DeVita and Hortobagyi, 2000; Lim et al., 2013; Silder et al., 2008). For each combination of step length and step frequency, visual indicators were fixed to the ground to constrain step length and a metronome was used to prescribe step frequency. Participants were given adequate practice time to walk as naturally and comfortably as possible while placing their feet close to the visual indicators and matching the cadence prescribed by the metronome beat. Each participant then performed five walking trials for each combination of step length and step frequency in a randomized order, and the errors between the experimental and prescribed step length and step frequency were calculated for each trial. The trial with the lowest error was chosen for further analysis. Overall, the error was less than 5% across all selected trials.

Kinematic, force-plate and muscle electromyography (EMG) data were recorded simultaneously for each walking trial during the stance phase of gait (heel-strike to toe-off).

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All data were collected from the subject's right lower limb, henceforth referred to as the ipsilateral limb. A three-dimensional (3D) video motion capture system (VICON, Oxford Metrics Ltd., Oxford) with nine cameras sampling at 120 Hz was used to record the 3D positions of small retro-reflective markers (14 mm in diameter) attached to each participant. Marker locations were chosen in accordance with those prescribed previously by Dorn et al. (2012b). Marker trajectories were filtered using a fourth-order Butterworth low-pass filter with a cut-off frequency of 4 Hz. Two strain-gauged force plates (Advanced Mechanical Technology Inc., Watertown, MA) mounted flush to the laboratory floor were used to measure ground reaction forces. Surface electrodes (Motion Laboratory Systems, Baton Rouge, LA) were attached to the right lower limb to record activity from six muscles: GMAX, GMED, vastus medialis, lateral hamstring, medial GAS and SOL. Onset and offset muscle activation times were determined by applying the Teager-Kaiser energy operator to the muscle-EMG data (Li et al., 2007; Solnik et al., 2010) and were used to qualitatively compare the sequence and timing of model-predicted muscle forces against measured EMG. Ground-reaction-force and muscle-EMG data were sampled at 1,080 Hz.

Calculations of lower-limb muscle forces were based on a generic musculoskeletal model developed by Anderson and Pandy (2001). The body was modelled as a 10-segment, 21-degree-of-freedom skeleton actuated by 54 muscle-tendon units. Each muscle-tendon unit was represented by a Hill-type muscle in series with an elastic tendon. Participant-specific musculoskeletal models were created by scaling segment lengths, segment inertial properties and muscle attachment sites in the generic model according to an individual participant's anthropometry.

Lower-limb muscle forces were determined using inverse dynamics and static optimization. At each instant of the gait cycle, an optimal set of joint angles was computed by

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minimizing the sum of the squares of the differences between the positions of virtual markers defined in the musculoskeletal model and reflective markers mounted on the participant. Net moments exerted about the joints were computed by inputting the optimal joint angles and measured ground forces to the model. Individual muscle forces were determined by minimizing the sum of the squares of muscle activations subject to the physiological bounds imposed by each muscle's force-length-velocity properties (Anderson and Pandy, 2001).

At each time step the vertical and fore-aft components of the ground reaction force vector were decomposed into contributions by individual muscles and gravity using a pseudoinverse-based approach described in detail by Lin et al. (2011) (see also Dorn et al. (2012a)). Briefly, five foot-ground interaction points were defined on the foot segment of the model to simulate the distributed force applied to the foot during ground contact. Each of these points could be fully-constrained, partially-constrained or unconstrained in space depending on the phase of foot-ground contact. At each time step during the stance phase the calculated joint angles were applied to the model. Each individual muscle force was then applied in isolation and a weighted least-squares optimisation problem solved to calculate the muscle-induced forces at each of the foot-ground interaction points. This procedure was then repeated for gravity. The contributions of individual muscle forces to the vertical and fore-aft components of the ground reaction force are hereafter described as 'vertical support' and 'forward progression', respectively. The contribution of all gravitational forces is referred to as 'limb posture' as this quantity describes the resistance to the downward pull of gravity provided by the bones and joints of the stance limb (Anderson and Pandy, 2003; Pandy and Andriacchi, 2010).

Nine different combinations of step length and step frequency were investigated (Fig. 1). Multiple regression analysis was used to determine and compare the individual effects of

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step length and step frequency on the following dependent variables: peak joint angles, peak joint moments, peak muscle forces, normalized muscle-fiber lengths, and peak muscle contributions to vertical support and forward progression. Normalized muscle-fiber lengths were analyzed at the times corresponding to peak muscle forces. Regression coefficients were calculated to determine the sensitivity of the dependent variables to changes in step length and step frequency. A significance level of $p < 0.05$ was assumed for all multiple regression analyses and only those correlations with a correlation coefficient of $R > 0.5$ are presented and discussed below.

RESULTS

Sagittal plane joint motion correlated significantly with changes in step length and step frequency at all walking speeds ($R>0.5$; $p<0.05$) (Fig. 2). Peak hip flexion, hip extension and knee flexion angles correlated more closely with changes in step length (Hip flexion: Coef =6.54; Hip extension: Coef =5.71; Knee flexion: Coef =6.61) than step frequency (Hip flexion: Coef =0.18; Hip extension: Coef =-0.17; Knee flexion: Coef =2.94). Peak hip extension, hip flexion, knee extension and ankle plantarflexion moments correlated significantly with changes in step length and step frequency ($R>0.5$; $p<0.05$) (Fig. 2). Peak knee extension and peak ankle plantarflexion moments correlated more closely with changes in step length (Knee extension: Coef =0.28; Ankle plantarflexion: Coef =0.16) than step frequency (Knee extension: Coef =0.19; Ankle plantarflexion: Coef =0.07) during first and second halves of stance, respectively. However, peak hip extension moment during the first half of stance correlated more closely with changes in step frequency (Hip extension: Coef =0.46) than step length (Hip extension: Coef =0.21).

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There was qualitative agreement between the sequence and timing of model-predicted muscle forces and EMG activity measured for SOL, GAS, VAS, HAMS, GMED and GMAX (Supplementary Fig. 3). Peak forces developed by SOL, VAS, HAMS, GMAX and ILPSO (iliacus and psoas major combined) correlated significantly with changes in both step length and step frequency ($R>0.5$; $p<0.05$) (Fig. 3 and Table 1). Step length had a greater effect on the peak forces developed by VAS and GMAX (VAS: Coef =7.62; GMAX: Coef =1.31) than step frequency (VAS: Coef =4.43; GMAX: Coef =0.63), whereas the peak forces developed by SOL, HAMS, and ILPSO were more substantially affected by step frequency.

Normalized muscle-fiber lengths at which peak hip and knee muscle forces were developed correlated more significantly with changes in step length and step frequency compared to the normalized fiber lengths at which peak ankle muscle force was developed ($R>0.5$; $p<0.05$) (Table 1). The normalized fiber lengths at which peak VAS, GMAX and ILPSO forces were developed were more sensitive to changes in step length (VAS: Coef =0.06; GMAX: Coef =0.05; ILPSO: Coef =0.05) than step frequency (VAS: Coef =0.03; GMAX: Coef =0.01; ILPSO: Coef =0.00) (Fig. 4 and Table 1).

The major hip, knee and ankle extensors, GMAX, GMED, VAS, SOL and GAS, dominated vertical support irrespective of changes in either step length or step frequency (see summed contributions of these muscles in Supplementary Fig. 7). Peak contributions of the hip and knee extensors to vertical support correlated more significantly with step length and step frequency than the peak contributions of the ankle muscles ($R>0.5$; $p<0.05$) (Fig. 5 and Table 1). Peak contributions of VAS, HAMS, and GMAX to vertical support during the first half of stance correlated more closely with changes in step length (VAS: Coef=2.29; HAMS: Coef =1.12; GMAX: Coef =0.83) than step frequency (VAS: Coef =1.33; HAMS: Coef =0.59; GMAX: Coef =0.29). Walking with either a larger step length or a higher cadence

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increased the summed contributions of all the stance-leg muscles to vertical support during the first half of stance while the contribution of limb posture to vertical support decreased. However, the effect of a larger step length was more pronounced than that of a higher cadence (ALLMUS 1st half: SL Coef=2.80 vs. SF Coef=1.43; ALLMUS 2nd half: SL Coef=1.24 vs. SF Coef=0.89; GRAV 1st half: SL Coef=-0.48 vs. SF Coef=-0.16).

The same five muscles also dominated forward progression irrespective of changes in either step length or step frequency (Supplementary Fig. 7). The backward-directed force generated by VAS correlated significantly with step length and step frequency during the first half of stance ($R>0.5$; $p<0.05$) (Fig. 6 and Table 1). Compared to step frequency, increasing step length resulted in greater increases in the contribution of VAS and in the summed contributions of all the stance-leg muscles to the braking force generated in the fore-aft direction during the first half of stance (VAS: SL Coef=-1.02 vs. SF Coef=-0.52; ALLMUS 1st half: SL Coef=-0.91 vs. SF Coef=-0.41). However, an increase in either step length or step frequency resulted in a smaller contribution from limb posture to the braking force generated during the first half of stance. The summed contribution of all stance-leg muscles to the forward acceleration of the center of mass correlated more closely with step length than step frequency during the second half of stance (ALLMUS 2nd half: SL Coef=0.85 vs. SF Coef=0.14).

DISCUSSION

We examined the separate effects of step length and step frequency on the individual contributions of the lower-limb muscles to vertical support and forward progression for a wide range of walking speeds in healthy young adults. We found that the same five muscles – GMAX, GMED, VAS, GAS, and SOL – dominated vertical support and forward progression

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independent of changes made to either step length or step frequency and that, overall, changes in step length had a greater effect on lower-limb joint motion, net joint moments and muscle function than step frequency. In support of our hypothesis, we also found that step length resulted in larger contributions from the hip and knee muscles and smaller contributions from gravitational forces (limb posture) to vertical support.

Our findings on the effects of changes in step length and step frequency on joint motion are in accordance with results reported by others. Sawicki and Ferris (2009) found that increasing step length leads to increases in hip extension, hip flexion and knee flexion. Winter (1983) found that participants walked with similar hip, knee and ankle joint motion when walking speed was increased by increasing step frequency. Liu et al. (2008), John et al. (2012), Jansen et al. (2013), and Pandy and Andriacchi (2010) showed that the primary hip, knee and ankle muscles – GMAX, GMED, VAS, GAS, and SOL – dominate vertical support and forward progression when humans are free to select any combination of step length and step frequency across the full range of walking speeds. These results are consistent with those of the present study.

The normalized fiber lengths of VAS and GMAX at which peak muscle forces were developed were more sensitive to a change in step length than step frequency (Fig. 4 and Table 1). Increasing step length caused an increase in both the hip and knee flexion angles during the first half of stance (Fig. 2 and Supplementary Fig. 1), allowing VAS and GMAX to operate at more favorable regions of their force-length relationships in the model (Fig. 4). In contrast, increasing step frequency did not alter the regions of the force-length relationships on which VAS and GMAX operated. These findings indicate the potential role of step length in generating higher VAS and GMAX forces and hence increased contributions to vertical support (Figs 3 and 5, Table 1). The contributions of VAS and GMAX to vertical

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support have been reported to increase with walking speed when step length and step frequency are freely selected (Liu et al., 2008; Pandy and Andriacchi, 2010). The results of the present study indicate that the contributions of the hip and knee muscles to vertical support during the first half of stance depend more heavily on step length than step frequency irrespective of walking speed.

Older adults walk more slowly by taking shorter steps and using a similar cadence to healthy young adults (Elble et al., 1991; Judge et al., 1996; Kerrigan et al., 1998; Winter et al., 1990). Our results indicate that longer step lengths require higher VAS and GMAX forces compared to a higher step frequency. We also found that the contribution of limb posture to vertical support was greater when participants walked more slowly by decreasing step length rather than step frequency. Older adults generally have weaker hip and knee extensors than healthy controls (Bohannon, 1997; Kerrigan et al., 1998). Our findings suggest that older adults may walk with a reduced step length to ease the muscular demand placed on VAS and GMAX and to exploit the increasing vertical support generated by limb posture. Whilst this result ought to be interpreted with caution as age-related changes in muscle strength were not explicitly modeled here, previous studies have demonstrated that model predictions of lower-limb muscle forces and muscle function in walking are relatively insensitive to changes in peak isometric muscle force (Ackland et al., 2012; Redl et al., 2007; Scovil and Ronsky, 2006).

In agreement with Jansen et al. (2013) and Liu et al. (2008) we found that limb posture provides less vertical support as walking speed increases (Table 1, Fig 6; and Supplementary Material). Our results also show that a reduced contribution from limb posture to vertical support is more prominent when walking speed is increased by increasing step length rather than step frequency. This finding may explain why older adults with

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reduced muscle strength walk faster by increasing their step frequency rather than taking longer steps (Elble et al., 1991). Taken together, our results imply that older adults with weaker muscles may rely more heavily on limb posture for vertical support compared to healthy controls, and that this dependence on limb posture is independent of walking speed.

There are limitations of the present study which should be recognized. First, each participant used a prescribed step length and step frequency at each walking speed. Although the participants were instructed to walk as naturally as possible, the effect of prescribing step length and cadence on gait biomechanics is unclear. Previous studies have used a similar approach to examine the effects of changes in step length and step frequency on the gait patterns of healthy individuals (Laurent and Pailhous, 1986; Martin and Marsh, 1991; Umberger and Martin, 2007). Furthermore, the values of step length and step frequency together with the hip, knee and ankle joint angles adopted by our participants were similar to those reported previously by others (Bertram, 2005; Bertram and Ruina, 2001; Sawicki and Ferris, 2009; Winter, 1983). Second, a cost function that minimized the sum of the squares of muscle activations was used to calculate lower-limb muscle forces when step length and step frequency were altered independently. It is possible that other criteria such as the cost of transport (Anderson and Pandy, 2001) may be equally, if not more well suited to examining the effects of changes in step length and step frequency on lower-limb muscle function during gait. However, previous studies have shown that the sum of the squares of muscle activations yields reasonable agreement between predicted muscle forces and measured EMG activity for a wide range of step lengths and step frequencies during walking (Dorn et al., 2012b; Liu et al., 2008; Pandy and Andriacchi, 2010). We also observed good temporal agreement between the calculated muscle forces and measured EMG activity for the participants in this study (Supplementary Fig. 3), suggesting that a minimum muscle-activation-squared criterion may

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be reasonable for modeling the effects of step length and step frequency on gait biomechanics in healthy individuals.

This is the first study to our knowledge to evaluate the separate effects of step length and step frequency on lower-limb muscle function during gait. Although increases in both step length and step frequency required larger contributions from the forces developed by GMAX, GMED, VAS, GAS and SOL to both vertical support and forward progression, an increase in step length resulted in greater differences in the contributions of VAS, GMAX and limb posture to vertical support. Previous studies have demonstrated that exercise programs can improve the walking abilities of older adults (Cadore et al., 2013). The present study provides additional insight into why older people with weak hip and knee extensors may walk more slowly by choosing different step lengths and step frequencies. Our results also aid in identifying the key muscle groups that may be targeted in exercise programs designed to improve gait biomechanics in elderly persons.

ACKNOWLEDGMENTS

This work was supported by a Discovery Projects Grant from the Australian Research Council (DP160104366) and an Innovation Fellowship from the Victorian Endowment for Science, Knowledge and Innovation awarded to MGP.

CONFLICT OF INTEREST

The authors do not have any financial or personal relationships with other people or organizations that could inappropriately influence this manuscript.

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FIGURE CAPTIONS

Fig. 1: Data were collected for nine different combinations of step length (SL) and step frequency (SF) constituting six walking conditions: Short SL, Nominal SL, and Long SL; Slow SF, Nominal SF and Fast SF. Each walking condition includes experimental data collected for three different combinations of step length and step frequency. Each combination of step length and step frequency is represented by a black dot with the corresponding walking speed indicated below. Panel A illustrates the data associated with three different combinations of step length while Panel B depicts data associated with three different combinations of step frequency.

Fig. 2: Peak angular displacements of the hip, knee and ankle joints and peak net joint moments developed by the hip, knee and ankle for each of the six walking conditions investigated. 1st half, first-half of the stance phase of gait; 2nd half, second-half of the stance phase of gait. Positive (negative) angular displacements denote hip and knee flexion (extension) and ankle dorsiflexion (plantarflexion). Positive (negative) joint moments signify hip and knee extension (flexion) and ankle plantarflexion (dorsiflexion).

Fig. 3: Peak muscle forces generated for the six conditions of step length and step frequency investigated. Each bar chart represents a mean value while the error bars represent one standard deviation. Muscle forces are normalized by body mass. Muscle symbols are as follows: SOL, soleus; GAS, gastrocnemius; VAS, vasti; HAMS, semimembranosus, semitendinosus, and biceps femoris long head combined; GMAX, lateral portion of gluteus maximus; GMEDA, the anterior portion of gluteus medius;

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GMEDP, the posterior portion of gluteus medius; ILPSO, iliacus and psoas major combined. 1st half, first-half of the stance phase of gait; 2nd half, second-half of the stance phase of gait.

Fig. 4: Effects of step length (blue horizontal bars) and step frequency (red horizontal bars) on normalized muscle-fiber length at the peak force developed by three representative muscles in the model. The left and right ends of each horizontal bar represent the normalized muscle-fiber length calculated at the minimum and maximum step length and step frequency, respectively. The arrows indicate the direction of muscle-fiber length change. Step frequency had a negligible effect on the normalized muscle-fiber lengths of GMAX and ILPSO as indicated by the red dots in the diagram. Muscle symbols are VAS, vasti; GMAX, lateral portion of gluteus maximus; ILPSO, iliacus and psoas major combined.

Fig. 5: Peak contributions of individual muscles and limb posture to the vertical ground reaction force (vertical support) for the six conditions of step length and step frequency investigated. Each bar chart represents a mean value while the error bars represent one standard deviation. Muscle contributions to vertical support are normalized by body mass. Muscle symbols are as follows: SOL, soleus; GAS, gastrocnemius; VAS, vasti; HAMS, semimembranosus, semitendinosus, and biceps femoris long head combined; GMAX, medial and lateral portions of gluteus maximus combined; GMED, anterior and posterior portions of gluteus medius combined;ILPSO, iliacus and psoas major combined; ALLMUS-1st half, contributions of all stance leg muscles combined in the first half of stance; ALLMUS-

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2nd half, contributions of all stance leg muscles combined in the second half of stance; and GRAV, contribution of limb posture. 1st half, first-half of the stance phase of gait; 2nd half, second-half of the stance phase of gait.

Fig. 6: Peak contributions of individual lower-limb muscles and limb posture to the fore-aft ground reaction force (forward progression) for the six conditions of step length and step frequency investigated. Each bar chart represents a mean value while the error bars represent 1SD. Muscle contributions to forward progression are normalized by body mass. Muscle symbols are defined in the caption for Figure 5. 1st half, first-half of the stance phase of gait; 2nd half, second-half of the stance phase of gait.

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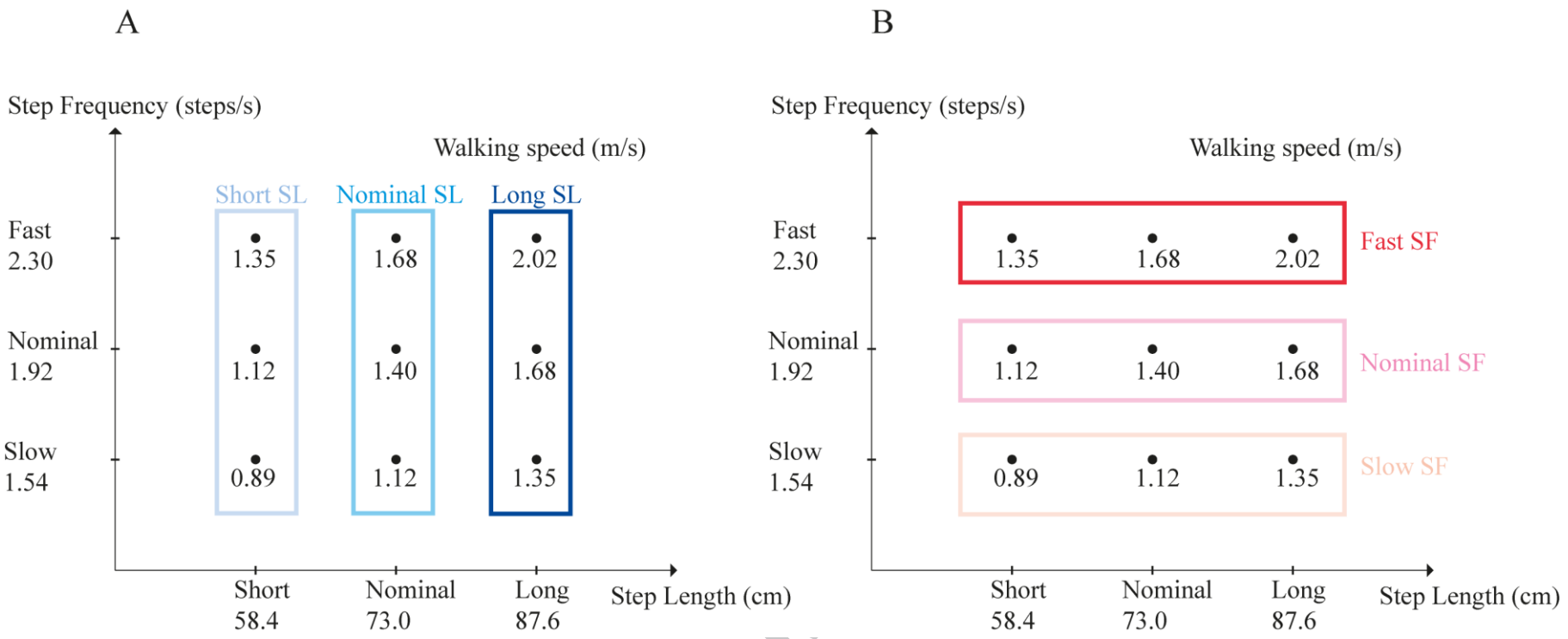


Figure 1

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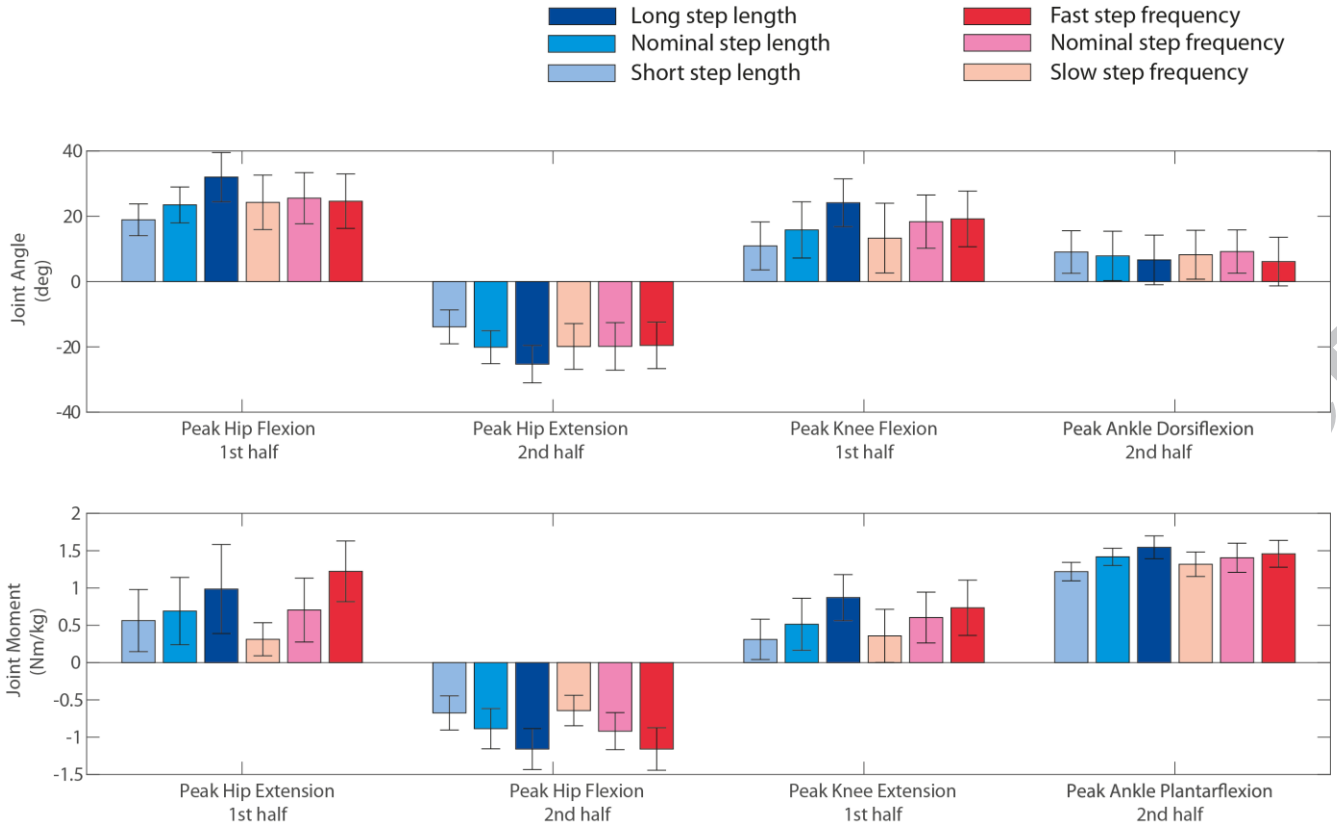


Figure 2

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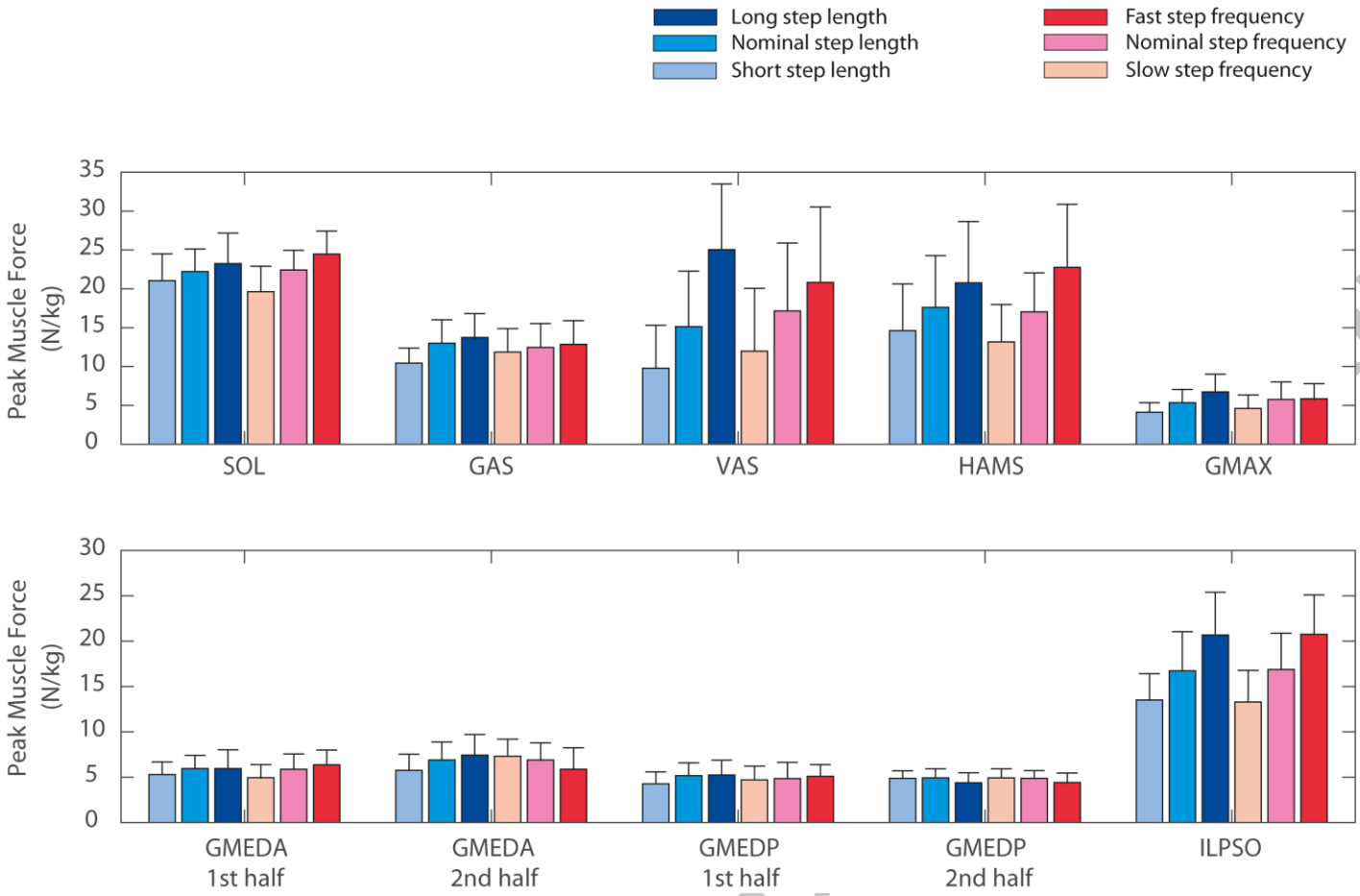


Figure 3

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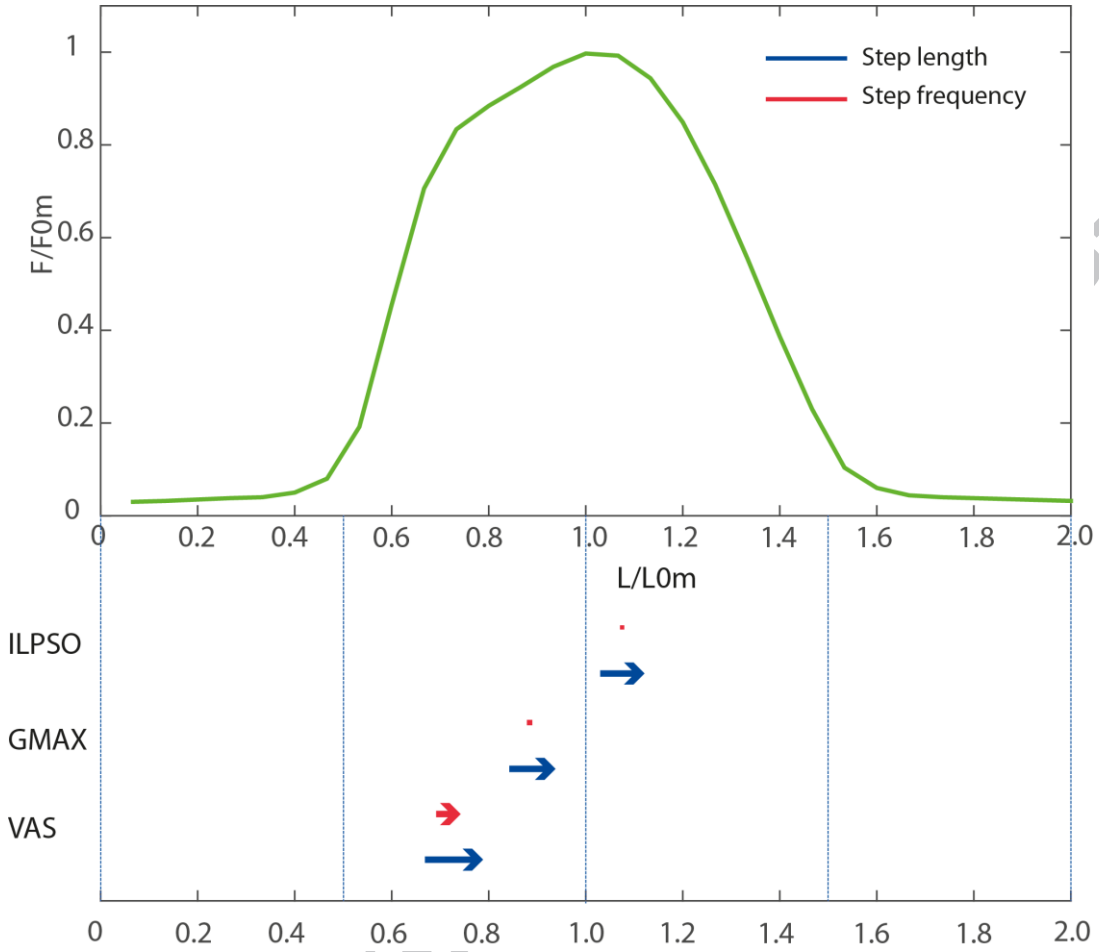


Figure 4

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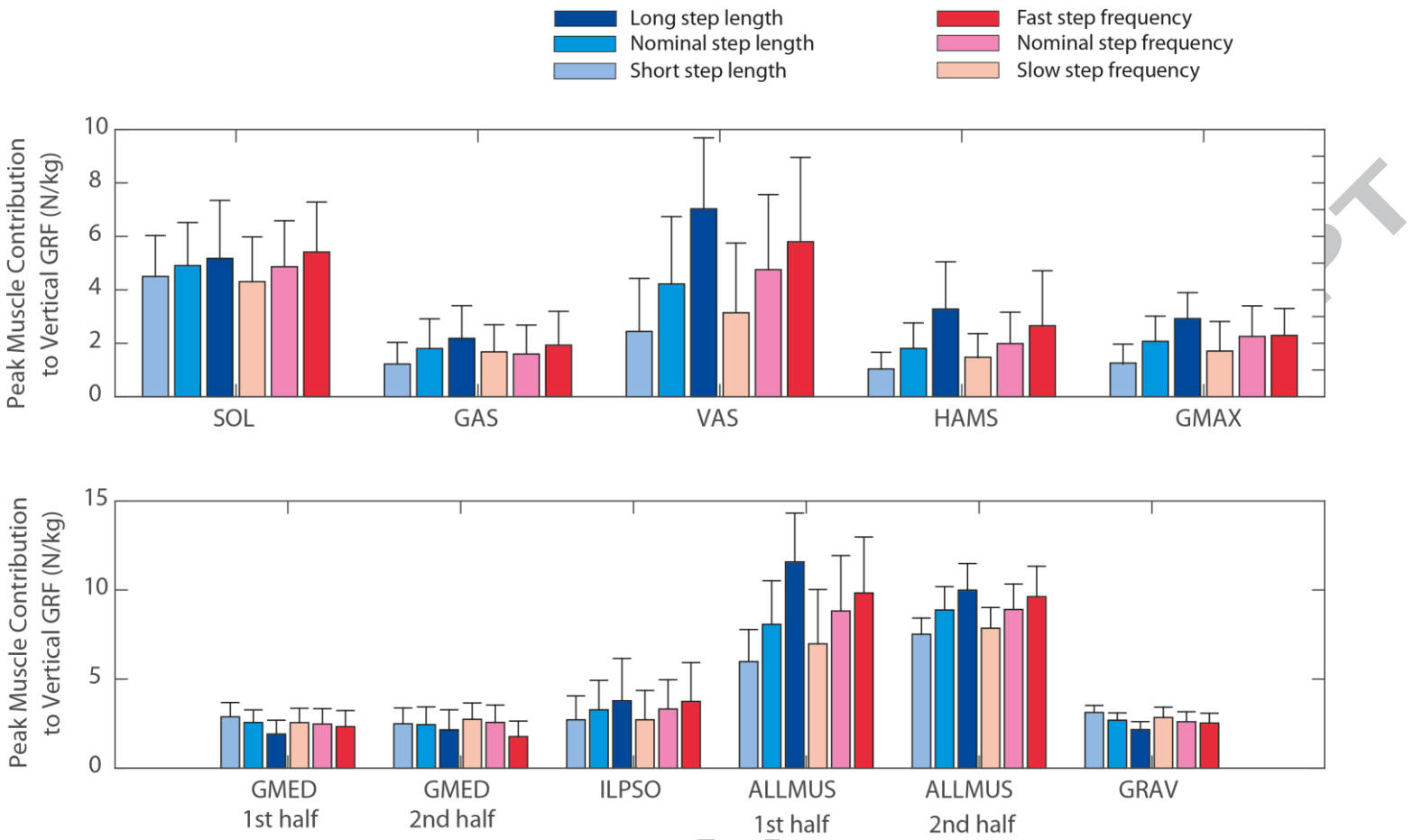


Figure 5

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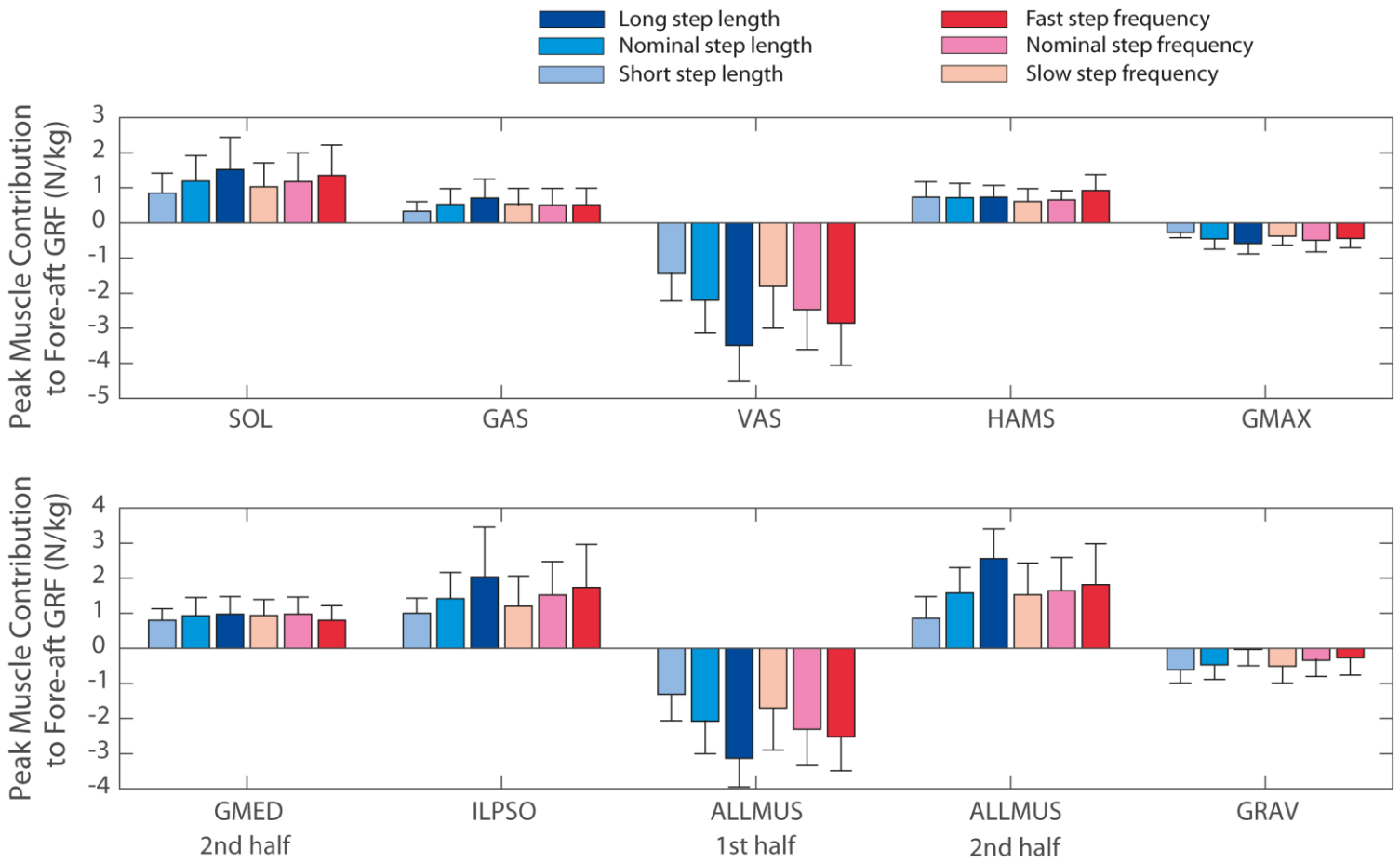


Figure 6

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Table 1: Results of the multiple regression analyses performed for peak muscle force, normalized fiber length at which peak muscle force is developed, and the peak contributions of individual muscles to the fore-aft and vertical ground reaction forces. The two independent variables are step length (SL) and step frequency (SF). R denotes the correlation coefficient while Coef denotes the regression coefficient. A negative correlation coefficient indicates that an increase in an independent variable would result in a decrease in a dependent variable. An asterisk denotes either a significant correlation or a significant regression coefficient with a significance level of $p < 0.05$. 1st half, first-half of the stance phase of gait; 2nd half, second-half of the stance phase of gait.

		Peak muscle force						Normalized fiber length at peak muscle force						Peak contribution to ground reaction force (vertical direction)						Peak contribution to ground reaction force (anterior-posterior direction)					
	Stance	R		SL Coef.		SF Coef.		R		SL Coef.		SF Coef.		R		SL Coef.		SF Coef.		R		SL Coef.		SF Coef.	
SOL	2nd half	0.62	*	1.09	*	2.43	*	0.28	*	-0.03	*	-0.02		0.30	*	0.34		0.56	*	0.39	*	0.34	*	0.17	
GAS	2nd half	0.46	*	1.64	*	0.50		0.17		0.01		-0.02		0.36	*	0.48	*	0.13		0.34	*	0.19	*	-0.01	
VAS	1st half	0.76	*	7.62	*	4.43	*	0.62	*	0.06	*	0.03	*	0.72	*	2.29	*	1.33	*	0.76	*	-1.02	*	-0.52	*
HAMS	1st half	0.64	*	3.07	*	4.80	*	0.47	*	0.03	*	-0.01		0.68	*	1.12	*	0.59	*	0.33	*	0.00		0.16	*
GMEDA	1st half	0.38	*	0.32		0.70	*	0.13		-0.01		0.00		-		-		-		-		-		-	
GMEDA	2nd half	0.43	*	0.85	*	-0.72	*	0.35	*	-0.02	*	-0.01		-		-		-		-		-		-	
GMEDP	1st half	0.28	*	0.48	*	0.19		0.47	*	0.02	*	0.01	*	-		-		-		-		-		-	
GMEDP	2nd half	0.29	*	-0.24		-0.26	*	0.41	*	-0.02	*	-0.02	*	-		-		-		-		-		-	
GMED	1st half	-		-		-		-		-		-		0.48	*	-0.48	*	-0.11		-		-		-	
GMED	2nd half	-		-		-		-		-		-		0.42	*	-0.17		-0.49	*	0.20		0.09		-0.07	
GMAX	1st half	0.58	*	1.31	*	0.63	*	0.67	*	0.05	*	0.01		0.65	*	0.83	*	0.29	*	0.47	*	-0.16	*	-0.03	
ILPSO	2nd half	0.85	*	3.58	*	3.73	*	0.62	*	0.05	*	0.00		0.33	*	0.53	*	0.52	*	0.46	*	0.52	*	0.27	*
ALLMUS	1st half	-		-		-		-		-		-		0.78	*	2.80	*	1.43	*	0.74	*	-0.91	*	-0.41	*
ALLMUS	2nd half	-		-		-		-		-		-		0.78	*	1.24	*	0.89	*	0.70	*	0.85	*	0.14	
GRAV	1st half	-		-		-		-		-		-		0.73	*	-0.48	*	-0.16	*	0.53	*	0.29	*	0.12	*

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		Peak muscle force				Normalized fiber length at peak muscle force				Peak contribution to ground reaction force (vertical direction)				Peak contribution to ground reaction force (anterior-posterior direction)			
	Stance	**R**		**SL Coef.**	**SF Coef.**	**R**		**SL Coef.**	**SF Coef.**	**R**		**SL Coef.**	**SF Coef.**	**R**		**SL Coef.**	**SF Coef.**
SOL	2nd half	0.62	*	1.09 *	2.43 *	0.28	*	-0.03 *	-0.02	0.30	*	0.34	0.56 *	0.39 *		0.34 *	0.17
GAS	2nd half	0.46	*	1.64 *	0.50	0.17		0.01	-0.02	0.36	*	0.48 *	0.13	0.34 *		0.19 *	-0.01
VAS	1st half	0.76	*	7.62 *	4.43 *	0.62	*	0.06 *	0.03 *	0.72	*	2.29 *	1.33 *	0.76 *		-1.02 *	-0.52 *
HAMS	1st half	0.64	*	3.07 *	4.80 *	0.47	*	0.03 *	-0.01	0.68	*	1.12 *	0.59 *	0.33 *		0.00	0.16 *
GMEDA	1st half	0.38	*	0.32	0.70 *	0.13		-0.01	0.00	-		-	-	-		-	-
GMEDA	2nd half	0.43	*	0.85 *	-0.72 *	0.35	*	-0.02 *	-0.01	-		-	-	-		-	-
GMEDP	1st half	0.28	*	0.48 *	0.19	0.47	*	0.02 *	0.01 *	-		-	-	-		-	-
GMEDP	2nd half	0.29	*	-0.24	-0.26 *	0.41	*	-0.02 *	-0.02 *	-		-	-	-		-	-
GMED	1st half	-		-	-	-		-	-	0.48	*	-0.48 *	-0.11	-		-	-
GMED	2nd half	-		-	-	-		-	-	0.42	*	-0.17	-0.49 *	0.20		0.09	-0.07
GMAX	1st half	0.58	*	1.31 *	0.63 *	0.67	*	0.05 *	0.01	0.65	*	0.83 *	0.29 *	0.47 *		-0.16 *	-0.03
ILPSO	2nd half	0.85	*	3.58 *	3.73 *	0.62	*	0.05 *	0.00	0.33	*	0.53 *	0.52 *	0.46 *		0.52 *	0.27 *

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ALLM US	1st half	-	-	-	-	-	-	0.78	*	2.80	*	1.43	*	0.74	*	-0.91	*	-0.41	*
ALLM US	2nd half	-	-	-	-	-	-	0.78	*	1.24	*	0.89	*	0.70	*	0.85	*	0.14	
GRAV	1st half	-	-	-	-	-	-	0.73	*	-0.48	*	-0.16	*	0.53	*	0.29	*	0.12	*